Introduction

For this project, we are simulating a tibia undergoing various conformational and force changes because of different scenarios. The bone is hollow with an inside diameter of 24.34 mm and an outside diameter of 25.1 mm and an overall length of 445 mm. This was determined from putting the last four digits of my RIN in an excel spreadsheet and getting the input and output dimensions as listed above. For this project, we are comparing the bone structure/composition of a young person's bone vs that of an older person's bone. The young and old bones have different physical material properties but, in this simulation, they will have the same base measurements as each other. In terms of general properties, the young bone has a higher Young's Modulus which means that it is generally stiffer than the older bone. This inherently causes some differences between young and old bones that we will delve further into.

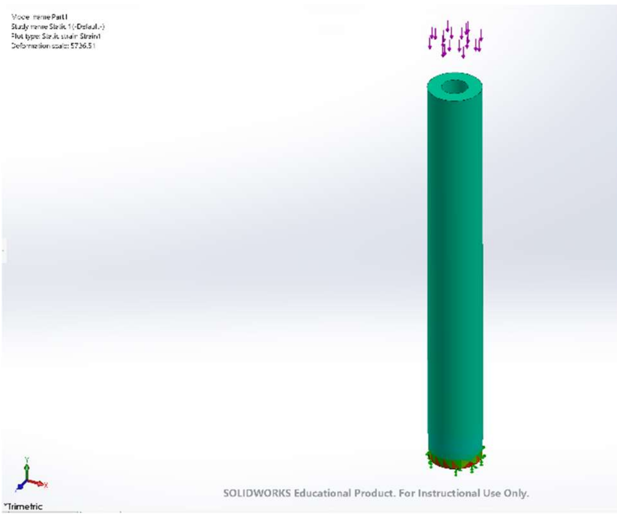
Part 1: Compression/Axial Loading

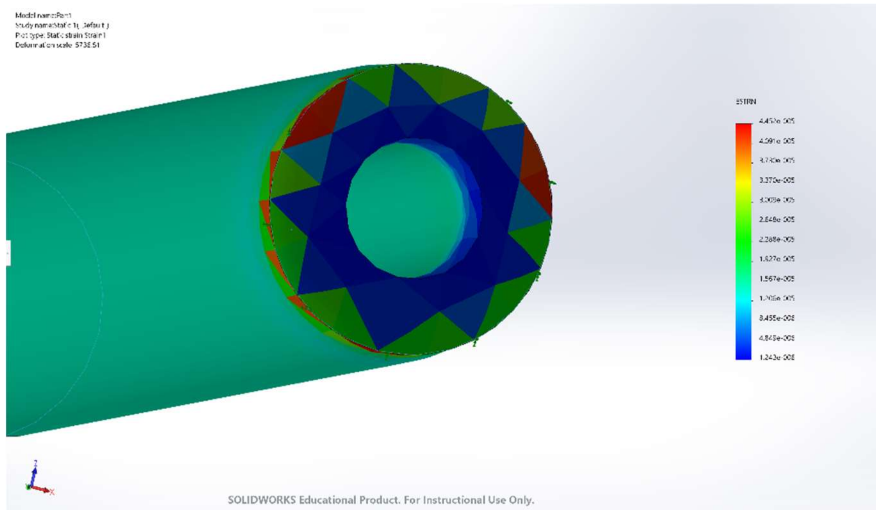
For this part, we are simulating a person's tibia as they are carrying a heavy object while walking. The information that is given to us is that there is a 400 N compression force acting on the top of the tibia while being fixed at the bottom. We are comparing how a compression force acts on the two types of bones as they change in material property as the bone progresses with age.

Simulations:

<u>Old Bone:</u>

Strain





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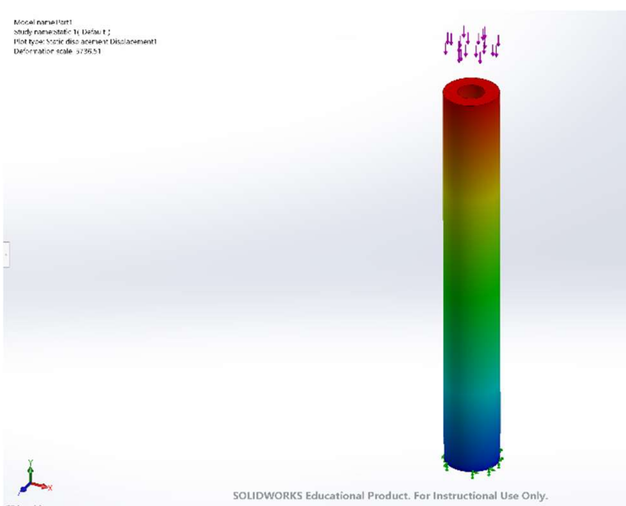
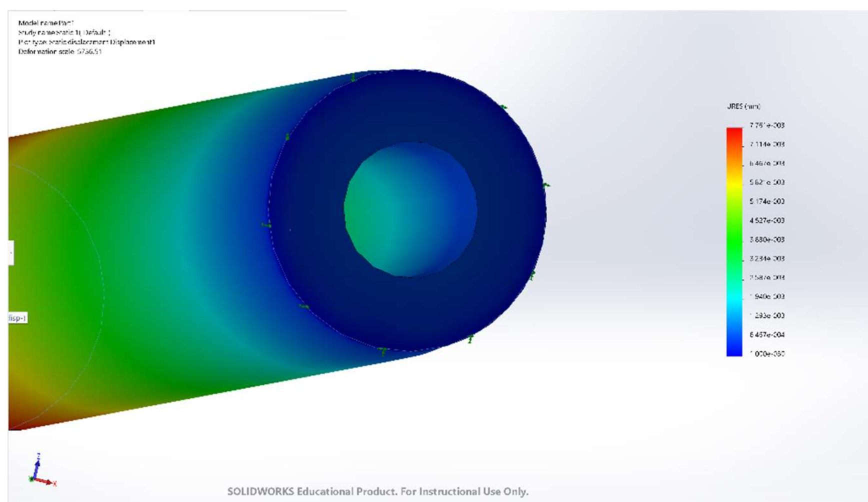
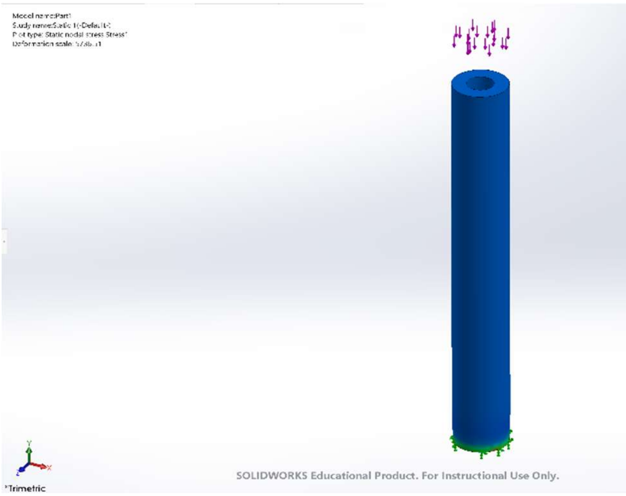
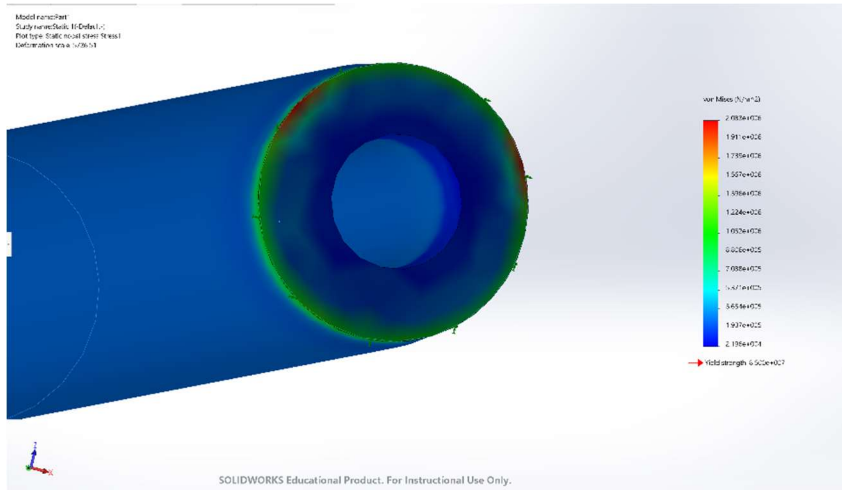




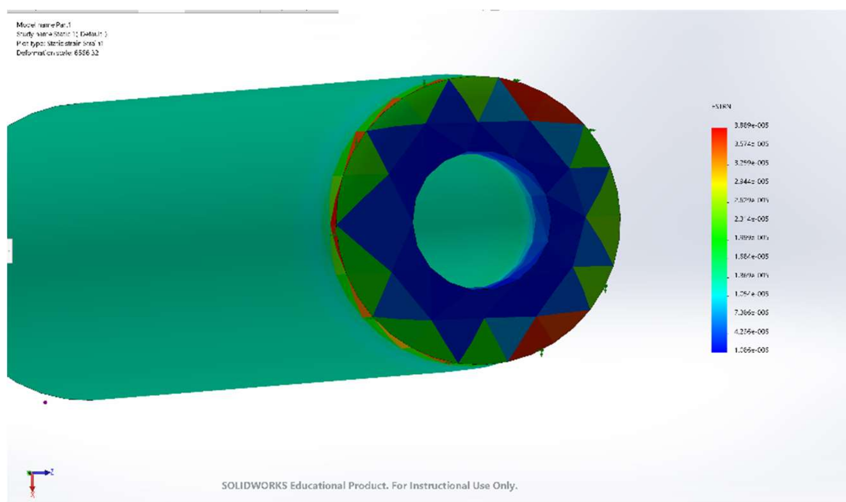
Stress:





<u>**Young Bone:**</u>

Strain:



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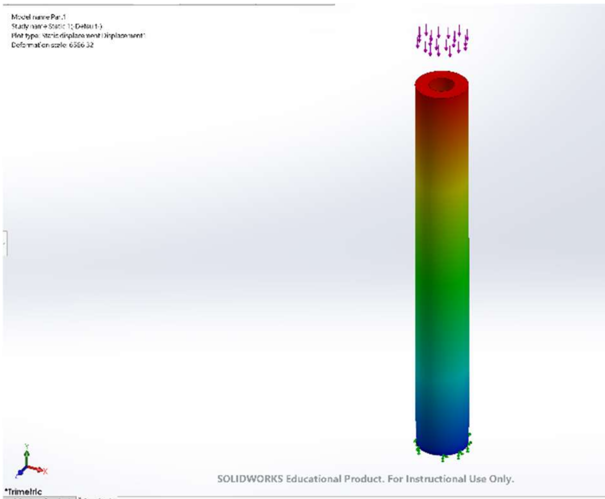
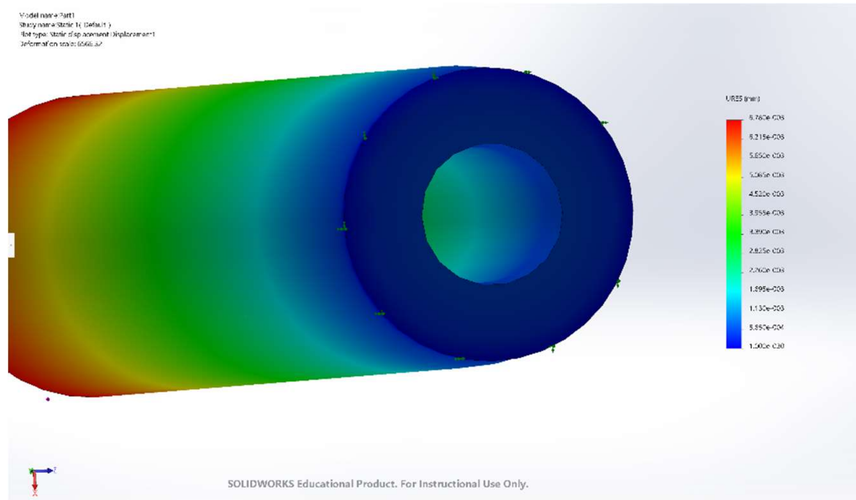
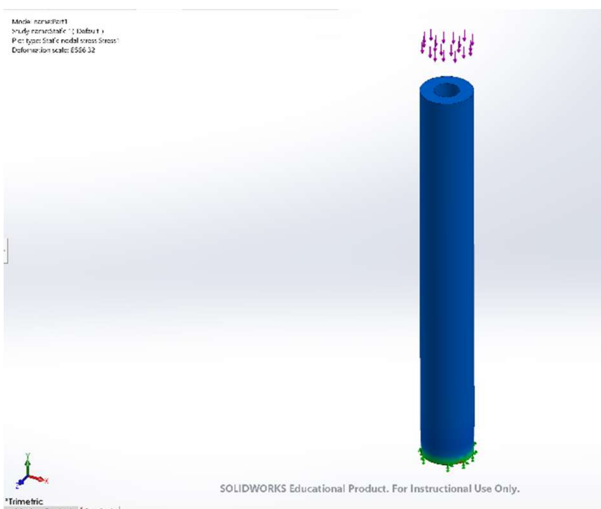
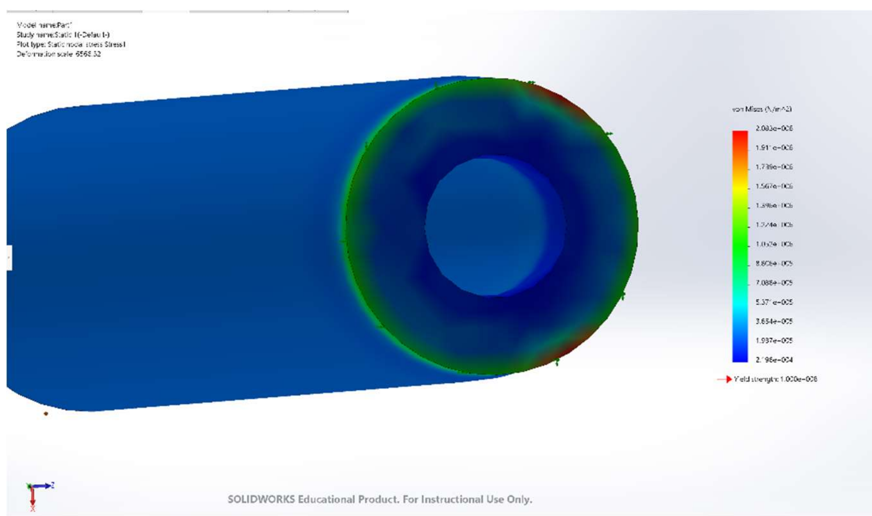




Stress:



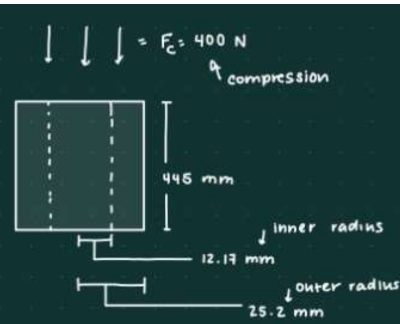


Calculations:

Stress / Contraction

$\sigma = \frac{F}{A}$ ← normal stress

$\epsilon = \frac{\sigma}{E}$ ← normal strain



Young Bone:

Normal Stress:

compression force

$$\sigma = \frac{F}{A} = \frac{-400\ N}{(\pi(.025m)^2)-(\pi(.01217m)^2)} = \frac{-400\ N}{1.498 \cdot 10^{-3}\ m^2} = \underline{-266987\ N/m^2}$$

total area

outside area

inside area

Normal Strain

$$\epsilon = \frac{\sigma}{E} = \frac{-266987\ N/m^2}{18.2 \cdot 10^9\ N/m^2} = \underline{-1.467 \cdot 10^{-5}}$$

↳ $E_{young} = 18.2\ GPa = 18.2 \cdot 10^9\ Pa = 18.2 \cdot 10^9\ N/m^2$

OLD BONE:

Normal Stress:

$$\sigma = \frac{F}{A} = \frac{-400\ N}{(\pi(.025m)^2)-(\pi(.01217m)^2)} = \frac{-400\ N}{1.498 \cdot 10^{-3}\ m^2} = \underline{-266987\ N/m^2}$$

Normal Strain

$$\epsilon = \frac{\sigma}{E} = \frac{-266987\ N/m^2}{15.8 \cdot 10^9\ N/m^2} = \underline{-1.700 \cdot 10^{-5}}$$

↳ $E_{old} = 15.8\ GPa = 15.8 \cdot 10^9\ Pa = 15.8 \cdot 10^9\ N/m^2$

Comparison of Calculations to Simulations:

When comparing the calculations to the simulations, it is evident that normal stress is the same for both the old bone and the young bone. On the calculations, the normal stress for both were equal to -266987 N/m^2 due to compressive forces but on the simulation, the normal stress was equal to 2.083e+006N/m^2 and undergoing compression. When comparing the strains however, the calculations and the simulation showed that the old bone had a greater value. In the hand calculation, the value for normal strain in the old bone, -1.7e-5, is 14.71% greater than the normal strain in the young bone which is -1.467e-5. When you compare these hand calculation values to the simulation values, you find that the simulation values are much higher than the hand calculations, with the calculations being around 78.31% less than the simulations.

Explanation:

From the different simulations we can make a few conclusions regarding stress and deformation of the two tibia bones. For one, the stress values for both simulations are the same meaning that despite having a difference in material properties of the two types of bones, they still undergo the same stress. In the pictures, it is evident that both bones have the same blue color regarding the stress diagram which means that they had similar stresses if not identical. This can also be proven with the equation since the stress equations for both young and old bones are the same.

The stress equations are the same because the values are not based on the individual properties of the bone (such as Young's Modulus or Shear Modulus) but rather the generalized physical properties that both bones have in common with each other. This means that in the equation for stress, the main components being used are the forces acting on the bone and the general area, which the bones have in common with each other. Therefore, both bones share the same stresses when presented with the same compression force and overall physical measurement values. The main difference between the two bones resides in deformation. In the deformation equation, the physical materials of each bone type have more of an impact on the bone itself. In the simulations above, it is noticeable that both bones have a spectrum of colors, indicating that deformation is happening at the same time across the bone at different magnitudes. This is because when there is a force of compression acting at the top of the tibia, the rest of the bone has to accommodate for the change by undergoing deformational changes. Like this, the different material properties of the bones play a part in how the bone deforms when undergoing compression. For example, the older bone undergoes more deformation due to it not being as dense and firm, therefore when faced with force, it gets affected more. This can also be proven by comparing the deformation calculations between the two types of bones as shown below. After looking at both simulations, it is evident that the young bone undergoes less deformation due to the higher young's modulus and compressive strength. It seems that as a bone matures with age, it becomes less dense and more "malleable" under a great amount of force, which in turn causes it to face deformation. Since the bone is sturdier and firmer in comparison to the old bone, when put under a great amount of force (such as 400 N) it doesn't change its shape as much. However, despite becoming less firm after a period, it seems as though the stress of the bone under a force doesn't change. This means that it stays relatively constant over the course of someone's lifespans.

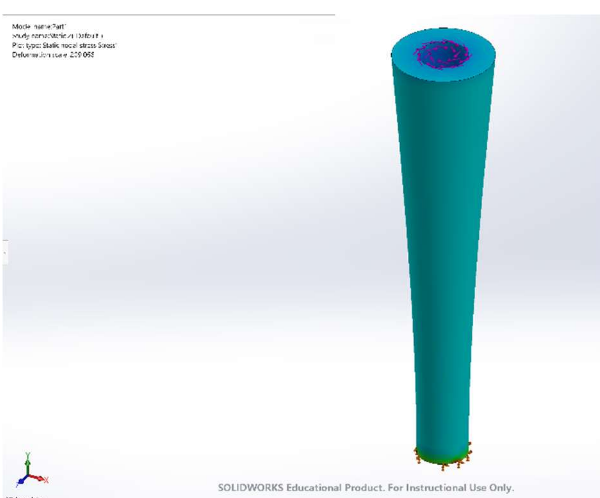
Part 2: Torque & 3-Point Bending

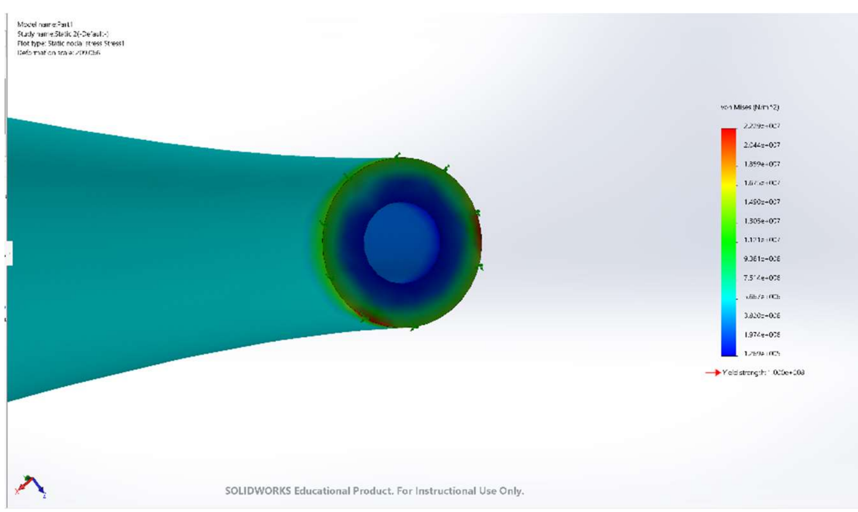
For this part, we are simulating a person's tibia as they are taking a step and pushing off the ground while simultaneously being hit by a large golden retriever on their knee. From this information, we know that the person had their tibia fixed to the ground and using the other information, we also know that there is a 75 Nm torque acting on the top of their tibia.

Simulation:

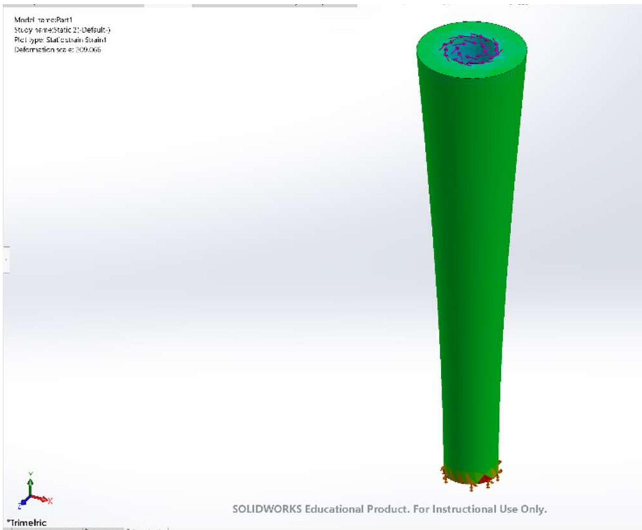
<u>Young Bone:</u>

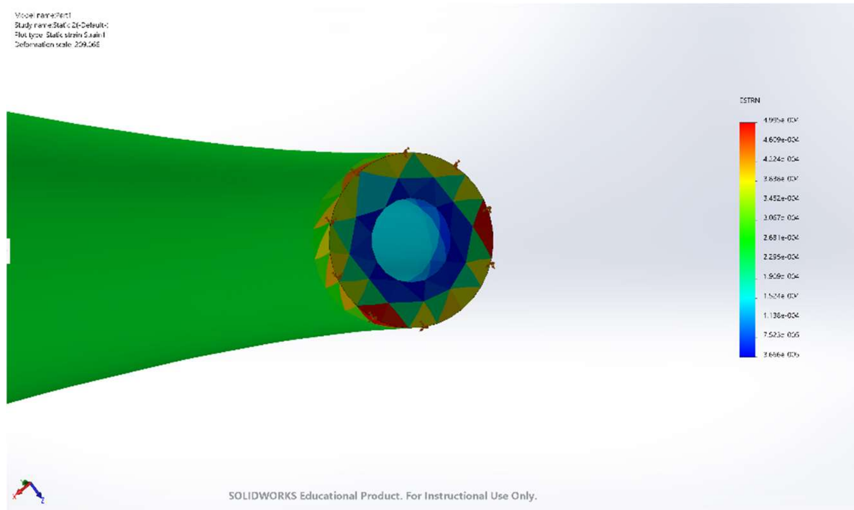
Stress:



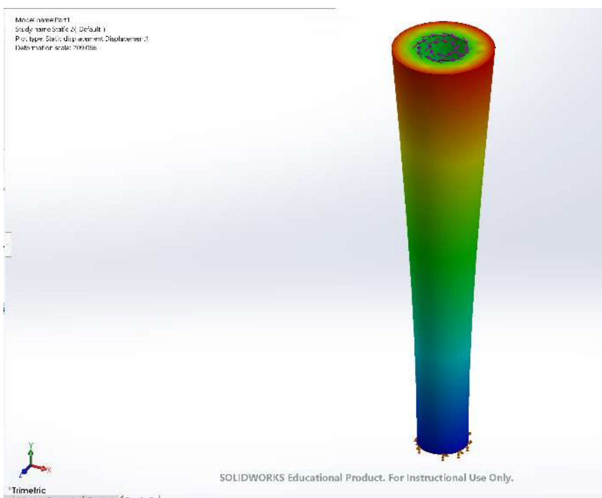


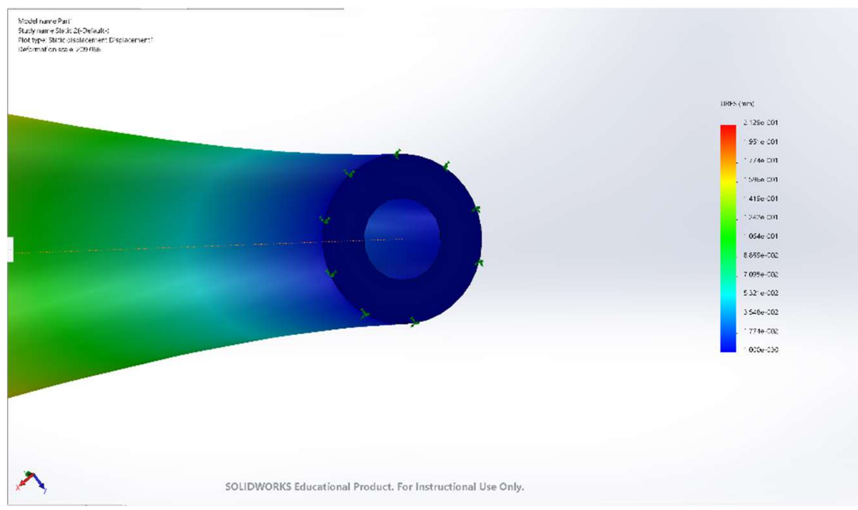
Strain:





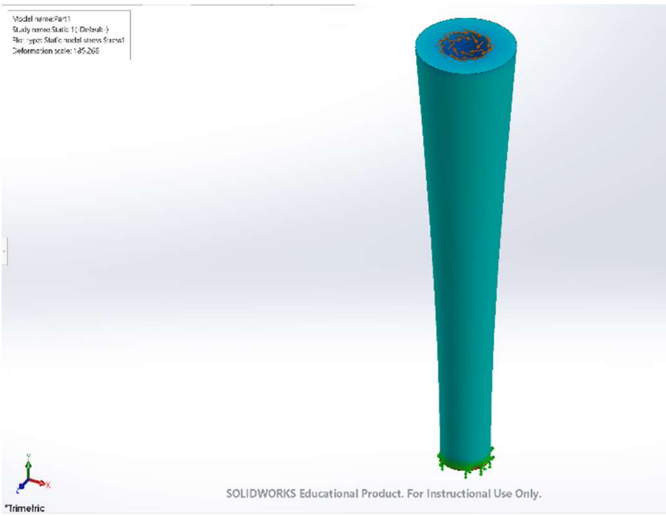
Displacement:

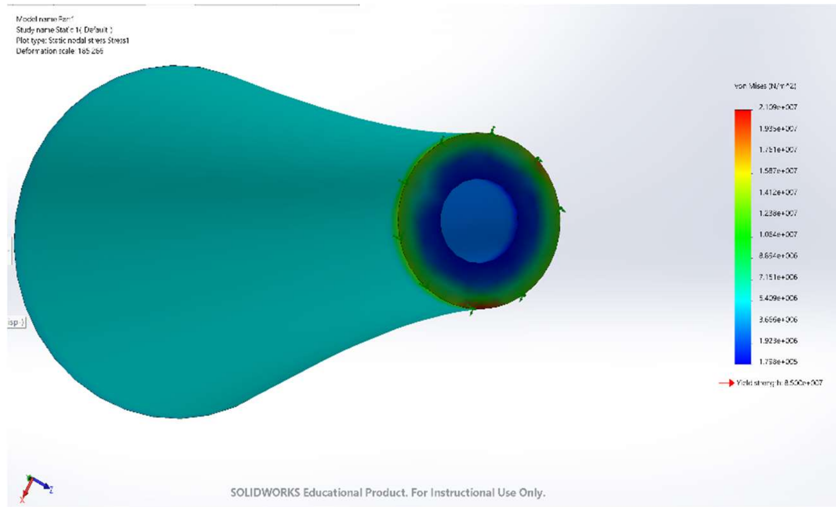




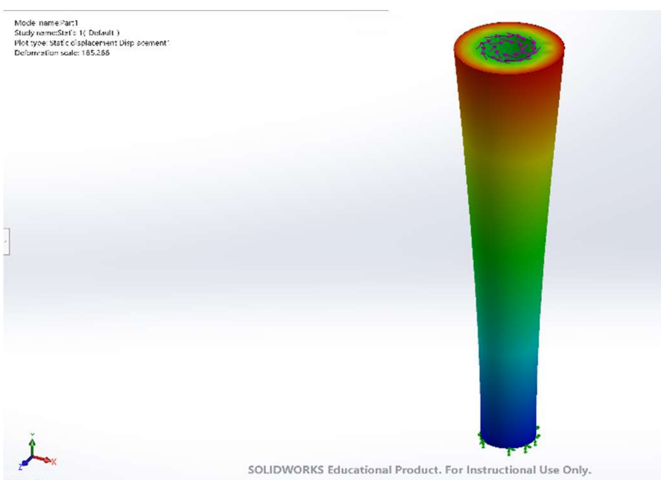
<u>**Old Bone:**</u>

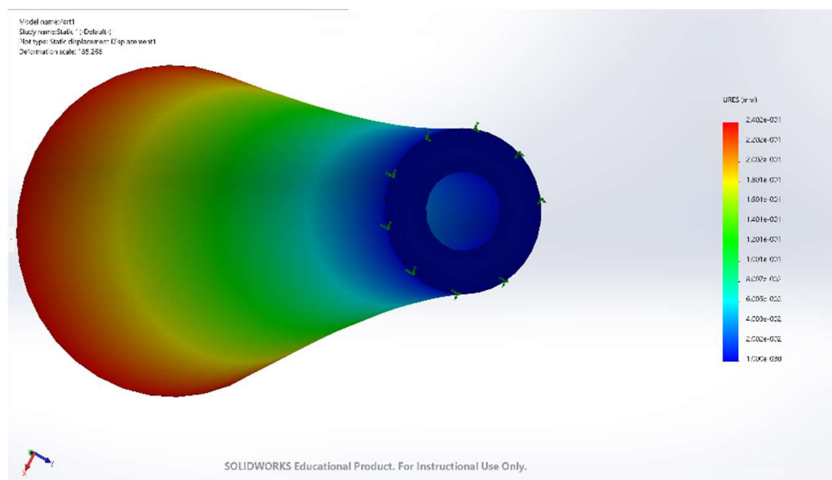
Stress:



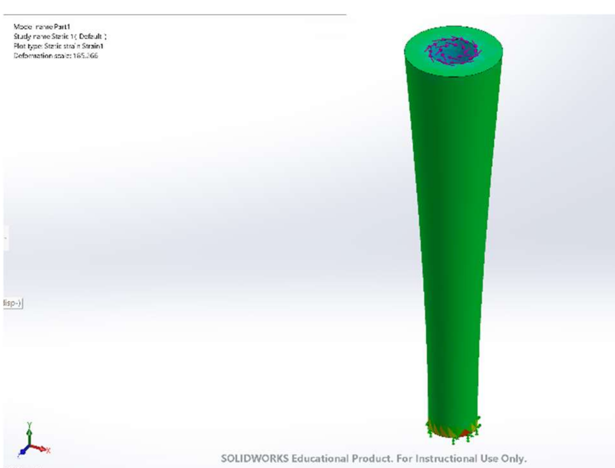


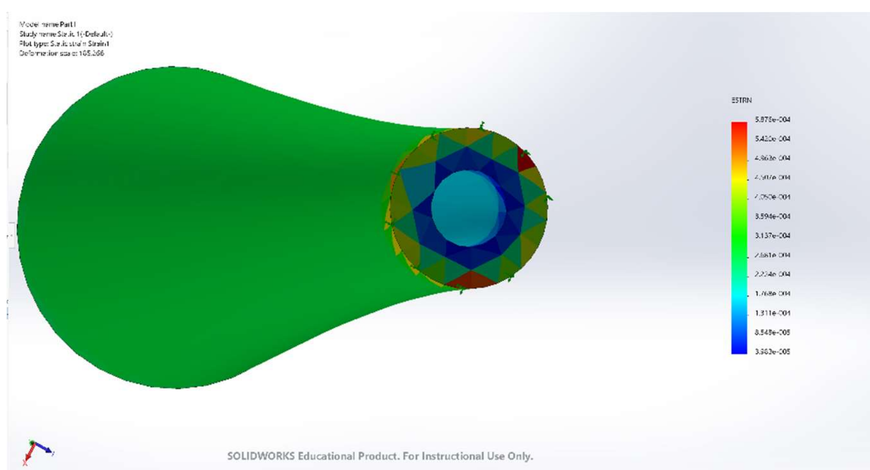
Displacement:



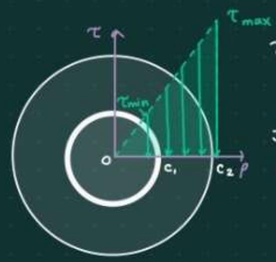


Strain:





Torque



τ = shearing stress

moment of inertia

$J = \frac{1}{2}\pi\left(c_2^4 - c_1^4\right)$

$\hookrightarrow J = \frac{1}{2}\pi\left((25.2\ mm)^4 - (12.17\ mm)^4\right)$

$= \frac{1}{2}\pi\left(403276\ mm^4 - 21936.2\ mm^4\right) = \frac{1}{2}\pi\left(381340\ mm^4\right)$

$= 599007\ mm^4 = 5.99007 \cdot 10^{-7}\ m^4$

inner radius $= c_1 = 12.17\ mm = .01217\ m$

outer radius $= c_2 = 25.2\ mm = .025\ m$

Length $= 445\ mm = .445\ m$

$T = 75\ N \cdot m$

Young Bone:

Shear Stress

$$\tau_{max} = \frac{Tc_2}{J} = \frac{(75\ N\cdot m)(.025\ m)}{5.99007 \cdot 10^{-7}\ m^4} = \boxed{3.13018 \cdot 10^6\ N/m^2}$$

$$\tau_{min} = \frac{Tc_1}{J} = \frac{(75\ N\cdot m)(.01217\ m)}{5.99007 \cdot 10^{-7}\ m^4} = \boxed{1.52377 \cdot 10^6\ N/m^2}$$

angle of twist: $\phi = \frac{TL}{JG} = \frac{(75\ N\cdot m)(.445\ m)}{(5.99007 \cdot 10^{-7}\ m^4)(4.5 \cdot 10^9\ Pa)} = \boxed{.012382^\circ = \phi}$

$G_{young} = 4.5\ GPa = 4.5 \cdot 10^9\ Pa$

Old Bone:

$$\tau_{max} = \frac{Tc_2}{J} = \frac{(75\ N\cdot m)(.025\ m)}{5.99007 \cdot 10^{-7}\ m^4} = \boxed{3.13018 \cdot 10^6\ N/m^2}$$

$$\tau_{min} = \frac{Tc_1}{J} = \frac{(75\ N\cdot m)(.01217\ m)}{5.99007 \cdot 10^{-7}\ m^4} = \boxed{1.52377 \cdot 10^6\ N/m^2}$$

angle of twist: $\phi = \frac{TL}{JG} = \frac{(75\ N\cdot m)(.445\ m)}{(5.99007 \cdot 10^{-7}\ m^4)(3.6 \cdot 10^9\ Pa)} = \boxed{.015477^\circ = \phi}$

$G_{young} = 3.6\ GPa = 3.6 \cdot 10^9\ Pa$

Explanation:

In a similar manner to the compression forces acting on the different bones, the stress caused by torque does not change as a person grows older. This can be proven using hand calculations (as shown below). For this problem, we are trying to find the maximum and minimum sheer stress values. Although this equation slightly differs from the stress equation from earlier, it ultimately follows the same principles. The sheer stress equation does not rely on material properties but rather the physical measurements of the bones themselves. As shown in the pictures above, the sheer stresses for both the young and the old diagrams both have a reddish color which means that they have a higher stress value spread across the respective bones.

When looking at the deformation, we are in a similar scenario as to the previous part. There is deformation occurring all along the sides of the bones but since the old bone has a lower Young's modulus, it undergoes much more deformation in comparison to the young bone. The highest value in the deformation key (located on the side in rainbow color) is 2.402 and it belongs to the old bone. When comparing the strain values between the old bone and the newer one, it differs slightly in pattern from the simulation in the previous part. In this simulation, there are lines of different colors going along the sides of the bone.

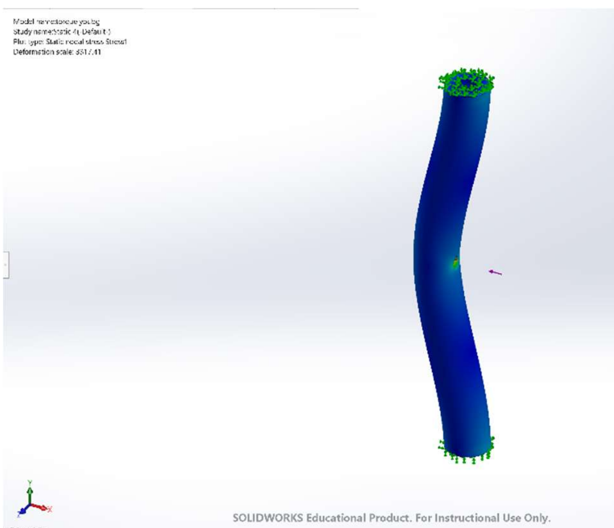
The second part of this section was to calculate the shear stress of the two bones respectively. After calculating, it is evident that the older bone undergoes a greater angle of twist in comparison to the younger bone. This is because in the equation for angle of twist, there is a Shears Modulus variable in the denominator. Since the older bone has a lower Shear modulus due to its increased fragility, it thus has a greater total angle of twist (as shown in the calculations below. It seems that when there is an external force acting on the bone, either torque or compression, the older bone is more likely to undergo more physical changes than the younger bone is. This is because when a person is young, their bones are more rigid and stiff, meaning that they can handle different forces without too much harm. However, the older a person gets, it seems that the more susceptible their bones are to deformation and twisting.

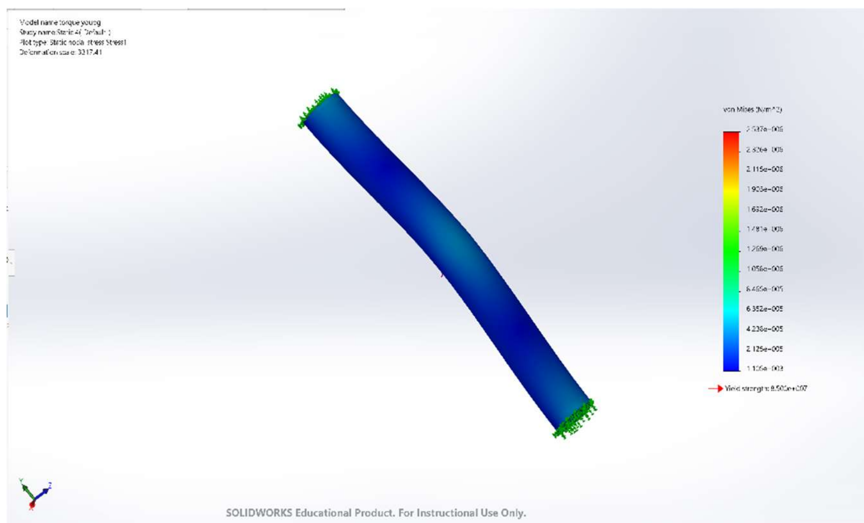
Part 3: 3-point Bending

For this last part, we are simulating a person's bone as they get hit in the midpoint of their leg while playing soccer. From this information, we know that the person had their tibia fixed to the ground and using the other information, we also know that there is a force of 100 N acting on the players midpoint of their tibia. The pictures below represent how an older bone might react in their situation in terms of stress and deformation.

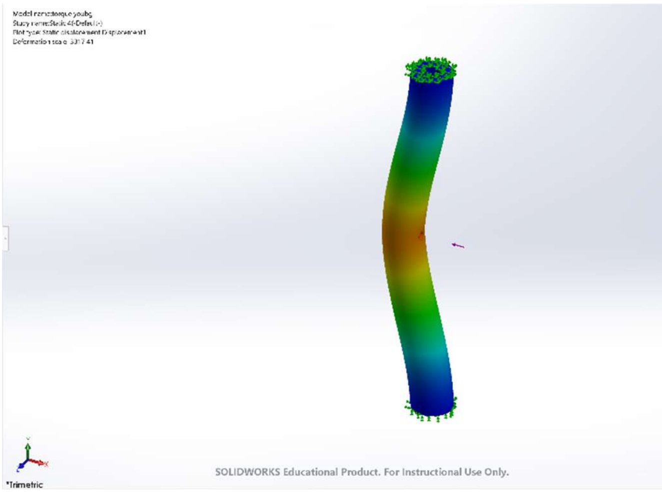
Simulation:

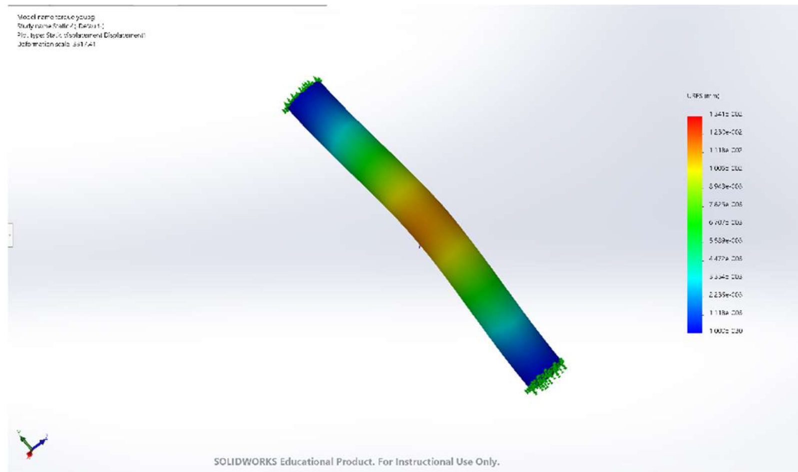
Stress:



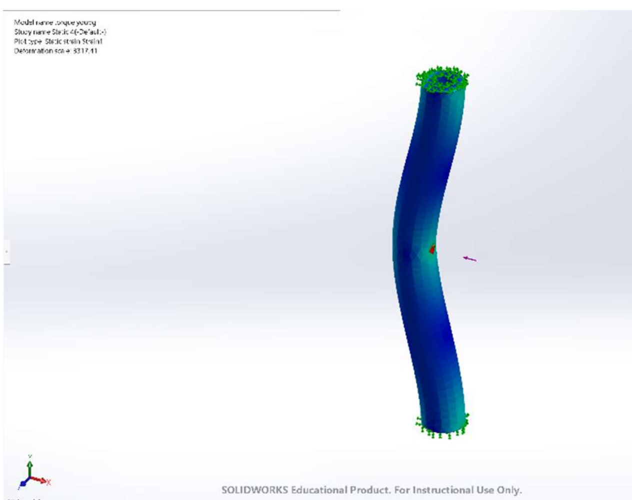


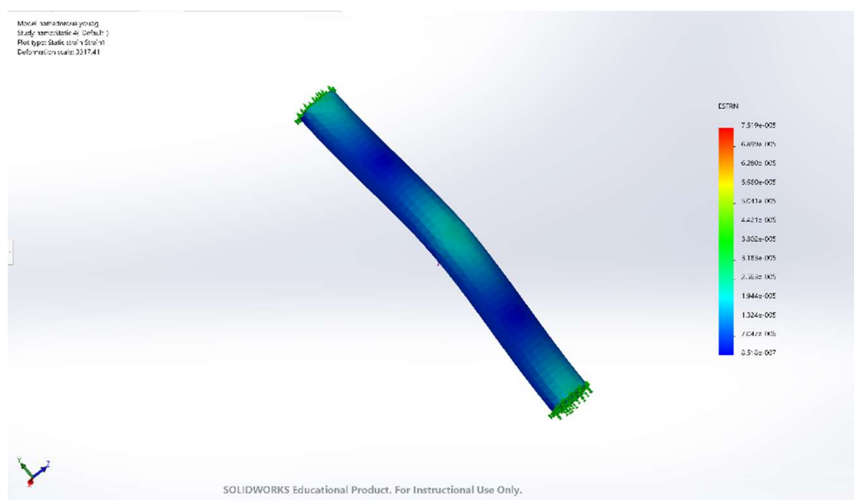
Displacement:



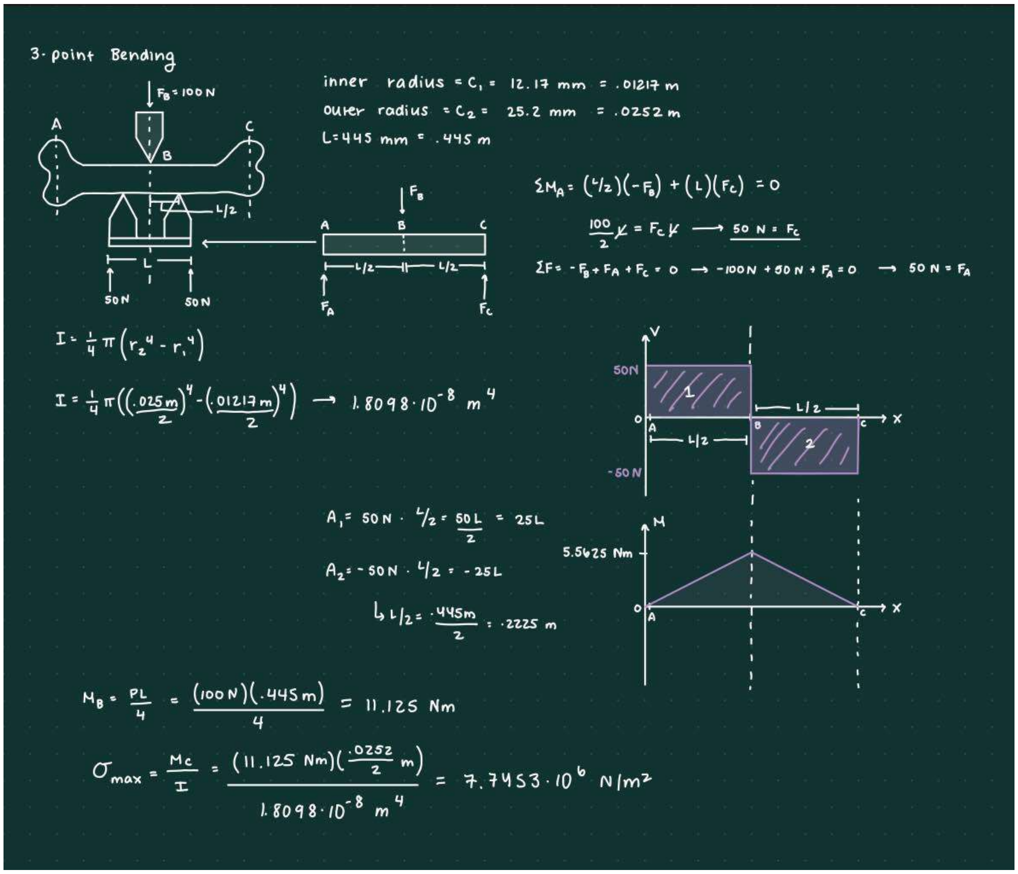


Strain:





Calculation:



Explanation:

From the pictures, we can see that there is a focus on stress coming from the midpoint of the bone. There is more of a reddish hue at the midpoint, showing that there is a higher amount of stress located there. Another focus point in the stress diagram is around the two fixed edges of the bone. While there is a high amount of stress located at the midpoint, there is also a relatively high amount of stress located at the two fixed ends of the bones. In a similar manner, when looking at the simulation correlated to deformation, the bone is most deformed at the midpoint/vertex where the force is acting. However, unlike stress and strain simulations, the deformation stimulation only has a high point at the midpoint of the bone, not the fixed sides as well. This is probably due to the force only acting at that point, leaving the other two ends to be forceless fixtures. Lastly, in the strain simulation, in a similar manner to the stress simulation, there is a focus on the midpoint and the two fixed ends. This is where the strain in the bone is the highest.

NOTE TO GRADER:

I apologize if the simulations are off, I followed the videos exactly, but my colors did not match theirs and my calculations didn't match my simulations despite doing similar calculations to those of other students. I additionally used similar calculations from when I took this course last Fall, and although my numbers were different, the overall trend in values differed as well.